

416 Knicks - API Contract

Jerry Zhu

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Summary

For Build 2510.3

Author: Jerry Zhu

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This document describes the API contracts for our backend to be consumed from the frontend, including the data structures, API endpoints and how to use them.

The API client library is in `frontend_client/src/api/client.ts` and should be used as the final reference.

To update this document, please install Pandoc and/or LaTeX and run the following command:

```
pandoc api-contract.md -f markdown --toc --toc-depth=4 --standalone -o api-contract.pdf
```

API Contracts

Data Structures

These are the data structures representing the JSON responses that the backend will return.

NOTE: This document only defines data structures, we define ourselves.

EAVS Data

`BallotStatisticsModel`

`MailBallotRejectionStatisticsModel`

`PollbookDeletionStatisticsModel`

`ProvisionalBallotStatisticsModel`

`VoterRegistrationStatisticsModel`

`ViewStateYearSummaryModel`

Geometry

All the data structures here relate to *EAVS* GeoUnit boundary data. Primarily centroids and the boundary polygons.

GeoUnitCentroidModel This is the data structure used to define the center of an *EAVS* GeoUnit, which is typically a county.

It is populated through as part of the preprocessing steps.

```
GeoUnitCentroidModel {
    fullRegionId: string; /// fully 10 digit padded fips code
    countyName: string;
    centerX: float; /// centerX and centerY are longitude and latitude.
    centerY: float;
}
```

GeoJSON We do not define this structure here, so please refer to the official GeoJSON Specification.

Voter Equipment

All the data structures here relate to voting equipment usecases.

VoterEquipmentModel This is the data structure used to define a voting equipment machine by make, type, and manufacturer.

It is based on the class-wide shared data spreadsheet, and is synced from it through a Python script.

This structure is currently read from a CSV file, instead of being retrieved from a database.

```
VoterEquipmentModel {
    manufacturer: string;
    equipmentType: string;
    modelName: string;

    // NOTE(jerry):
    // These are intentionally optional. We distinguish the state of
    // not knowing the value vs. a falsy value.

    discontinued: boolean?;
    firstManufactured: int?; // Year
    lastManufactured: int?; // Year
    operatingSystem: string?;

    vvpap: bool?; // Voter Verified Paper Audit Trail
```

```

        // whether the machine produces a paper receipt
        // of the users' digital ballot.
        //
        // Can be used for recounting.
    certificationLevel: string?;
    securityRiskDescription: string?;
}

```

Detailed Voter Registration

All the data structures defined here pertain to *Detailed Voter Registration* usecases.

VoterRegistrationDetailModel This is the data structure used to define a registered voter's information.

It contains enough to identify the person, and be used as part of a *Display Registered Voters* functionality.

```

VoterRegistrationDataModel {
    regionId: string;
    firstName: string;
    middleName: string;
    lastName: string;
    partyAffiliation: string;
    status: string;
}

```

Endpoints

These are the endpoints with their appropriate `client.ts` function name for invocation.

EAVS Data

GET /state/{fipsCode}/provisional-ballots

GET /state/{fipsCode}/{countyFipsCode}/provisional-ballots

GET /state/{fipsCode}/voter-registration-count

GET /state/{fipsCode}/{countyFipsCode}/voter-registration-count

GET /state/{fipsCode}/pollbook-deletions

GET /state/{fipsCode}/{countyFipsCode}/pollbook-deletions

GET /state/{fipsCode}/mail-ballot-rejections

GET /state/{fipsCode}/{countyFipsCode}/mail-ballot-rejections

GET /state/{fipsCode}/year-summary

GET /state/{fipsCode}/year-summary/{year}

Geometry

GET /state/{fipsCode}/geometry

GET /state/{fipsCode}/centroids

Voter Equipment

Detailed Voter Registration